

# CS 639: Intro to Foundation Models

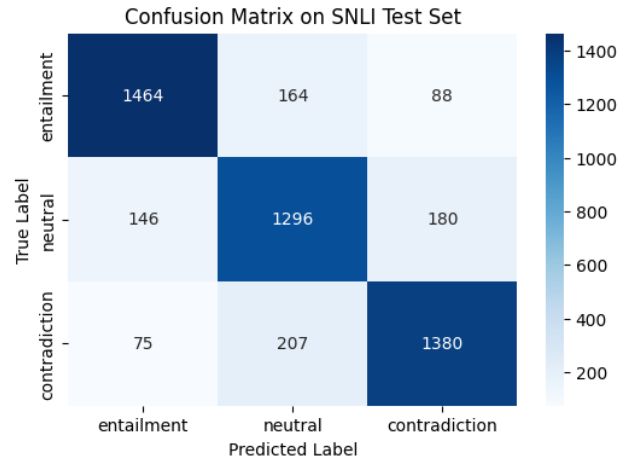
## Homework 4

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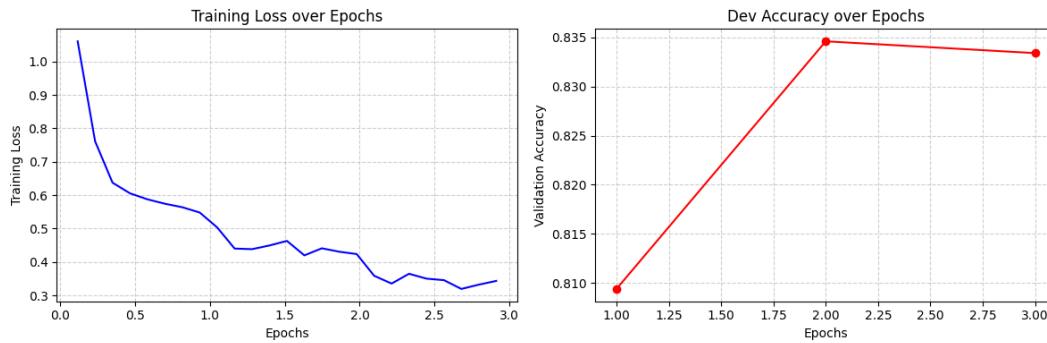
### Part 1: Baseline Full Fine-Tuning

(1.3) The test accuracy obtained was 82.80%.

The confusion matrix across all three labels was obtained as below -



The training loss and dev accuracy curves are as shown below -



Per-class precision, recall, and F1 scores for entailment, neutral, and contradiction is listed below -

| Class         | Precision | Recall | F1-Score |
|---------------|-----------|--------|----------|
| Entailment    | 0.87      | 0.85   | 0.86     |
| Neutral       | 0.78      | 0.80   | 0.79     |
| Contradiction | 0.84      | 0.83   | 0.83     |

(1.4) Based on the metrics, the model handles the entailment class best (highest F1-score: 0.86) and struggles most with the neutral class (lowest F1-score: 0.79).

The model most frequently confuses the neutral and contradiction classes. Linguistically, entailment is easier due to high word overlap between sentences. In contrast, both neutral and contradiction statements introduce entirely new words. Without real-world common sense, the model struggles to distinguish if these new details are simply unverified (neutral) or definitively false (contradiction).

(1.5)

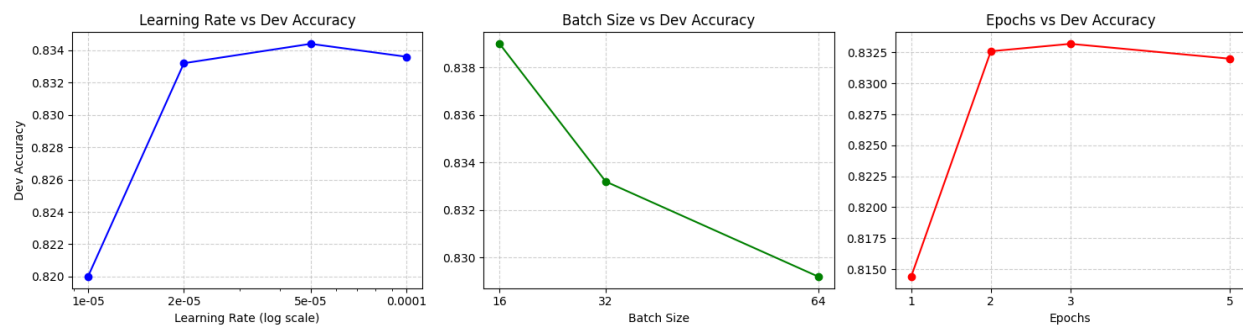
1. The model most frequently confuses the neutral and contradiction labels. Both classes introduce new words not found in the premise. Without real-world context, the model has a hard time telling unverified information (neutral) apart from impossible information (contradiction).
2. Yes, there are minor signs of overfitting. While the training loss keeps decreasing through the 3rd epoch, the dev accuracy peaks at the 2nd epoch and then slightly drops.
3. Entailment often shares matching words, and contradiction uses explicit negatives (like "not" or "never"). The neutral class requires actual logical reasoning, making it much more difficult.

## Part 2: Hyperparameter Sensitivity Study

(2.3) The table below lists every hyperparameter configuration tested and its corresponding dev accuracy. When varying one hyperparameter, all others were kept at their baseline values (from Task 1).

| Hyperparameter | Value Tested       | Dev Accuracy (%) |
|----------------|--------------------|------------------|
| Learning Rate  | $1 \times 10^{-5}$ | 82.00            |
|                | $2 \times 10^{-5}$ | 83.32            |
|                | $5 \times 10^{-5}$ | 83.44            |
|                | $1 \times 10^{-4}$ | 83.36            |
| Batch Size     | 16                 | 83.90            |
|                | 32                 | 83.32            |
|                | 64                 | 82.92            |
| Epochs         | 1                  | 81.44            |
|                | 2                  | 83.26            |
|                | 3                  | 83.32            |
|                | 5                  | 83.20            |

The figure below illustrates the relationship between each hyperparameter and the resulting dev accuracy.



(2.4) Based on how much the dev accuracy changed across the tested values, the hyperparameters can be ranked for their impact as follows:

1. Epochs: This caused the biggest increase. Accuracy jumped from 81.44% at 1 epoch to 83.31% at 3 epochs (an increase of 1.87%).
2. Learning Rate: This had the second biggest impact. Accuracy jumped from 82.00% at  $1e-5$  to a peak of 83.45% at  $5e-5$  (an increase of 1.45%).
3. Batch Size: This had the smallest overall swing. Accuracy dropped from 83.90% at a batch size of 16 to 82.91% at a batch size of 64.

The absolute best configuration tested was a batch size of 16, which achieved the highest overall accuracy of 83.90%. The worst configuration was training for only 1 epoch, which resulted in the

lowest accuracy of 81.44%. The gap between the best and worst settings is 2.46%.

Based on the results, the best setting should be one where the learning rate is  $5e - 5$ , the batch size is 16, and number of epochs is 3.

(2.5)

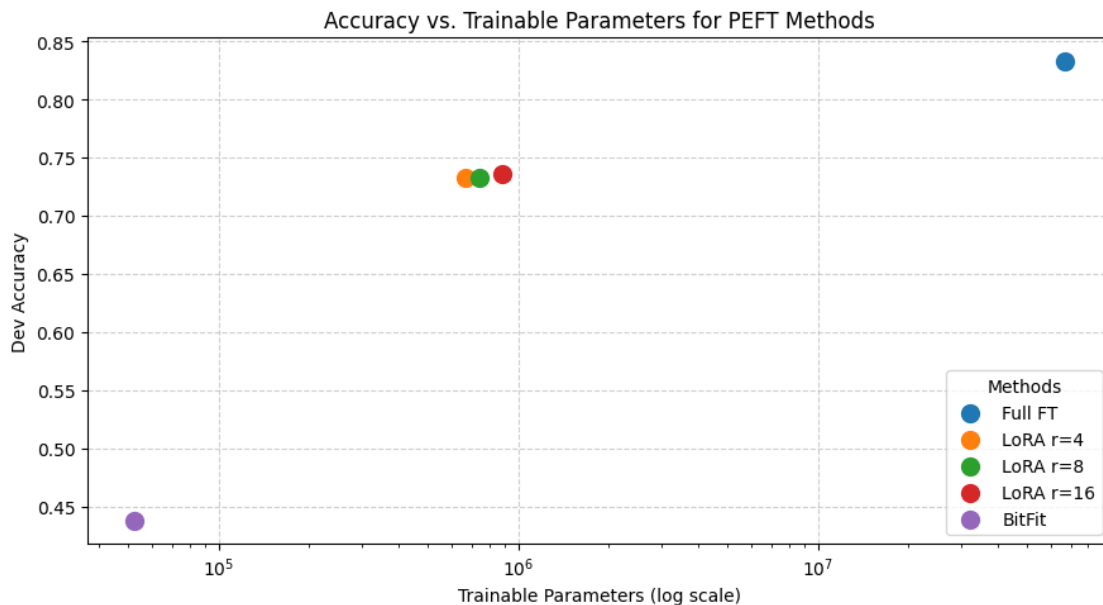
1. A learning rate that is too high causes the optimizer to take steps that are too large. Instead of smoothly converging to the minimum of the loss landscape, it overshoots, causing the loss to diverge instead of converging.
2. While a larger batch size gives a more stable gradient, the noise from smaller batch sizes acts as implicit regularization. This noise helps bump the model out of sharp local minima into flatter minima that generalize better to unseen data.
3. This indicates overfitting, where the model starts memorizing the training data instead of learning general patterns. To mitigate this, I would use early stopping to halt training when dev accuracy drops or increase regularization/dropout.

### Part 3: Parameter-Efficient Fine-Tuning (PEFT)

(3.3) The table below compares the trainable parameters, dev accuracy, and training time across full fine-tuning, LoRA at various ranks, and BitFit.

| Method      | Trainable Params | % of Total | Dev Acc (%) | Train Time (mins) |
|-------------|------------------|------------|-------------|-------------------|
| Full FT     | 66,955,779       | 100.00%    | 83.32       | 9.29              |
| LoRA (r=4)  | 666,627          | 0.99%      | 73.24       | 6.68              |
| LoRA (r=8)  | 740,355          | 1.09%      | 73.26       | 6.55              |
| LoRA (r=16) | 887,811          | 1.31%      | 73.56       | 6.56              |
| BitFit      | 52,227           | 0.08%      | 43.78       | 5.91              |

The figure below plots the dev accuracy against the number of trainable parameters for each method



(3.4) LoRA captures the vast majority of full fine-tuning performance even at a very small rank. At rank  $r = 4$ , it achieves roughly 73.24% accuracy. Increasing the rank to  $r = 8$  or  $r = 16$  provides almost no additional benefit (plateauing around 73.56%). This suggests that  $r = 4$  is the optimal rank for this specific task.

While BitFit trains an incredibly small number of parameters (only 0.08% of the total), its performance is significantly worse, achieving only 43.78% accuracy. LoRA offers a far superior trade-off; by training just slightly more parameters, it achieves nearly 30% higher accuracy than BitFit.

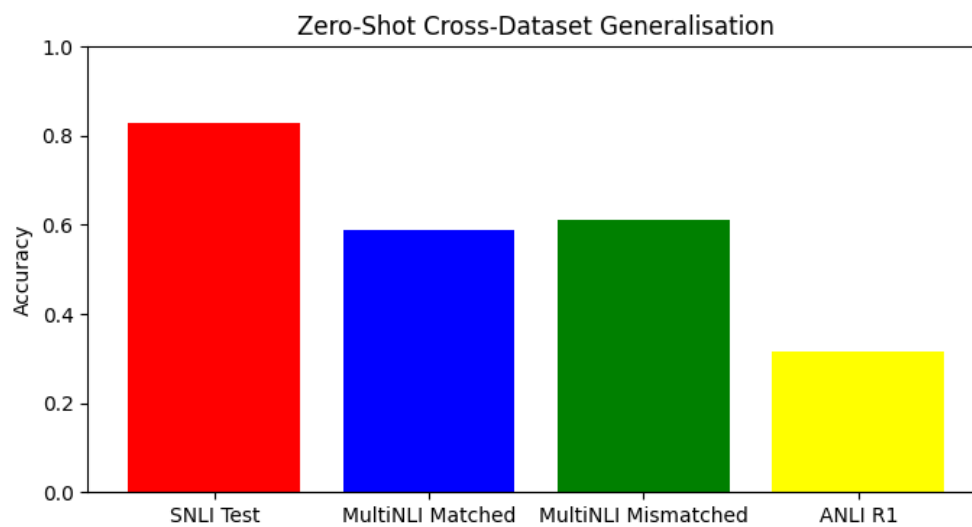
The decision to prefer a certain method in practical scenarios could be based on memory constraints. If GPU VRAM during training is limited, a very low-rank LoRA or BitFit is preferred, as they require storing drastically fewer optimizer states. On the other hand, in applications where primary bottleneck is not the training memory, but a different factor is a bigger concern, full fine-tuning should be preferred.

(3.5)

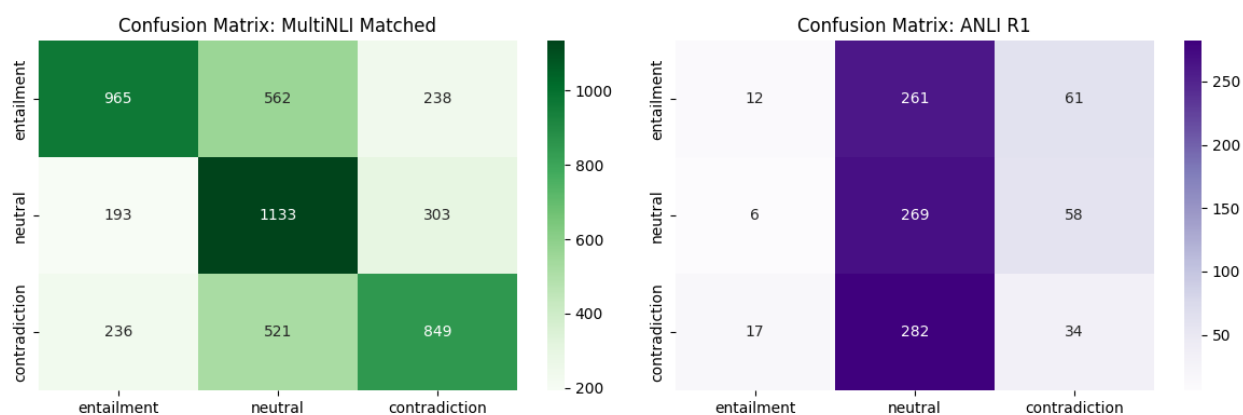
1. Pre-trained models already possess deep linguistic knowledge. Fine-tuning merely adapts this knowledge to a specific task format, meaning the optimal changes exist in a small mathematical subspace that a low-rank matrix can easily capture.
2. Biases act as simple shifts to activation thresholds. For NLI, the model often just needs to amplify or suppress existing pre-trained logical features. Tuning just the biases is an efficient way to adjust these thresholds without altering the core representations.
3. If increasing the LoRA rank yields no improvement, it suggests the overall task complexity is relatively low compared to the model's capacity.

## Part 4: Cross-Dataset Generalisation

(4.3) The bar chart below displays the zero-shot accuracy of the trained model across four different evaluation datasets.



The confusion matrices below illustrate the prediction distributions for the two datasets where the model performed the worst i.e. MultiNLI Matched and ANLI R1.



Below are 10 manually inspected misclassified examples from the worst-performing dataset (ANLI R1), along with a brief explanation for the failure:

- Premise:* Ernest Jones is a British jeweller and watchmaker. Established in 1949, its first store was opened in Oxford Street, London. Ernest Jones specialises in diamonds and watches, stocking brands such as Gucci and Emporio Armani. Ernest Jones is part of the Signet Jewelers group.

*Hypothesis:* The first Ernest Jones store was opened on the continent of Europe.

*True:* entailment | *Predicted:* neutral

*Reason:* The model lacks the geographic world knowledge that London is located in Europe.
- Premise:* Old Trafford is a football stadium in Old Trafford, Greater Manchester, England,

and the home of Manchester United. With a capacity of 75,643, it is the largest club football stadium in the United Kingdom, the second-largest football stadium, and the eleventh-largest in Europe. It is about 0.5 mi from Old Trafford Cricket Ground and the adjacent tram stop.

*Hypothesis:* There are only 10 larger football stadiums in Europe.

*True:* entailment | *Predicted:* contradiction

*Reason:* The model fails at the basic numerical reasoning required to understand that being eleventh means exactly ten are larger.

3. *Premise:* Magnus is a Belgian joint dance project of Tom Barman (from the rock band dEUS) and CJ Bolland. Magnus' debut album, "The Body Gave You Everything", was released on March 29, 2004. Two of its tracks, "Summer's Here" and "Jumpneedle", were released as singles.

*Hypothesis:* "The body gave you everything" album was not released on March 28, 2003 but on March 29, 2004.

*True:* entailment | *Predicted:* contradiction

*Reason:* The model is tricked by the explicit negation word "not" into automatically assuming a contradiction.

4. *Premise:* Takaaki Kajita ( 加賀 隆晃, Kajita Takaaki ) is a Japanese physicist, known for neutrino experiments at the Kamiokande and its successor, Super-Kamiokande. In 2015, he was awarded the Nobel Prize in Physics jointly with Canadian physicist Arthur B. McDonald.

*Hypothesis:* Arthur B. McDonald is a Japanese physicist, known for neutrino experiments at the Kamiokande and its successor, Super-Kamiokande.

*True:* contradiction | *Predicted:* neutral

*Reason:* The model struggles with complex syntax and fails to bind the correct nationality to the corresponding person.

5. *Premise:* Javier Torres (born May 14, 1988 in Artesia, California) is an undefeated Mexican American professional boxer in the Heavyweight division. Torres was the second rated U.S. amateur boxer in the Super Heavyweight division and a member of the Mexican Olympic team.

*Hypothesis:* Javier was born in Mexico

*True:* contradiction | *Predicted:* neutral

*Reason:* The model sees the word "Mexican" and fails to use the explicit California birthplace to definitively contradict the statement.

6. *Premise:* "How Could I Want More" is the debut single by American actress and singer Jamie Lynn Spears. The song was written by Spears and Rivers Rutherford, and was self-released (under the label name Sweet Jamie Music, Inc.) on November 25, 2013 as the lead single from Spears' debut EP, "The Journey".

*Hypothesis:* The title of Spears' debut EP was "Journeys."

*True:* contradiction | *Predicted:* neutral

*Reason:* The model likely ignores the minor morphological difference of the plural 's', treating the title as unverified rather than strictly false.

7. *Premise:* About to Crack is the fifth studio album by Dutch punk hardcore band Vitamin X. Released through Tankcrimes on September 11, 2012 in the US, Agipunk in Europe and Peculio Discos in Brazil. The album was recorded at Electrical Audio in Chicago by Steve

Albini who previously recorded Nirvana, Neurosis, PJ Harvey, High on Fire, Iggy Pop & The Stooges. Art is by John Dyer Baizley.

*Hypothesis:* Vitamin X made four albums before About to Crack.

*True:* entailment | *Predicted:* neutral

*Reason:* The model lacks the logical deduction skills to infer that a fifth album implies exactly four came before it.

8. *Premise:* Dyllan McGee is a documentary filmmaker and founder of McGee Media. In partnership with Peter Kunhardt, McGee produced "Gloria: In Her Own Words" (HBO), "Finding Your Roots with Henry Louis Gates, Jr." (PBS), "MAKERS: Women Who Make America" and many more. McGee is the Founder and Executive Producer of AOL's MAKERS.

*Hypothesis:* D McGee is a filmmaker who founded their own company. They made many films, that were broadcast on many US networks.

*True:* entailment | *Predicted:* neutral

*Reason:* The model lacks the external world knowledge required to know that HBO and PBS are US television networks.

9. *Premise:* The family of Macintosh operating systems developed by Apple Inc. includes the graphical user interface-based operating systems it has designed for use with its Macintosh series of personal computers since 1984, as well as the related system software it once created for compatible third-party systems.

*Hypothesis:* Macintosh is a business that owns Apple Inc.

*True:* contradiction | *Predicted:* neutral

*Reason:* The model falls for a lexical overlap trap where the exact same entities are present but their relationship is reversed.

10. *Premise:* Oksa is a village in Jedrzejów County, Świętokrzyskie Voivodeship, in south-central Poland. It is the seat of the gmina (administrative district) called Gmina Oksa. It lies approximately 18 km north-west of Jedrzejów and 41 km south-west of the regional capital Kielce. The village has an approximate population of 890.

*Hypothesis:* The population of Oksa is large

*True:* contradiction | *Predicted:* neutral

*Reason:* The model cannot evaluate the subjective, real-world concept of whether 890 people constitutes a "large" population.

(4.4) The accuracy drops massively on out-of-domain datasets. The in-domain SNLI accuracy is 82.80%, while out-of-domain drops to around 60% for MultiNLI and plummets to 31.50% for ANLI, representing a maximum performance gap of over 51%.

In this specific run, MultiNLI Matched (58.94%) and Mismatched (61.14%) performed very similarly. Typically, mismatched is harder because it contains text from genres completely unseen during the MultiNLI training phase, introducing new vocabulary and sentence structures.

Based on the 10 inspected ANLI errors, the failures fall into world knowledge required (examples 1, 8, 10), complex syntax and logical reasoning (examples 2, 4, 5, 7), and lexical overlap traps (examples 3, 6, 9). The complex syntax and world knowledge categories dominate the errors.

(4.5)

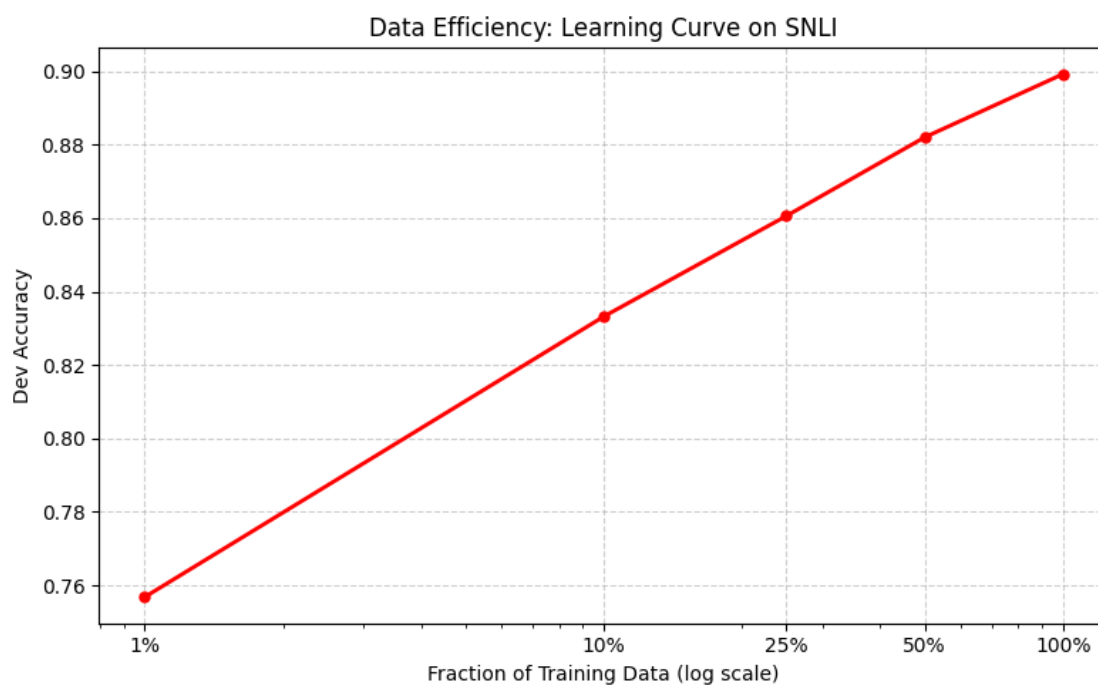
1. ANLI was constructed by humans to adversely fool NLI models, specifically targeting known weaknesses like lexical overlap traps and complex negation. MultiNLI is just naturally occurring text, which is less deceptive.
2. SNLI premises are image captions, which are structurally simple and highly literal. MultiNLI contains complex syntax, figurative language, and varied vocabulary from diverse genres. The model overfits to simple grammar and struggles to parse these more complex sentence structures.
3. Strategy 1: Train the model on a combination of datasets (like SNLI plus MultiNLI) to expose it to diverse genres and syntactic structures, preventing domain overfitting.  
Strategy 2: Include adversarial examples in the training data to penalize the model for relying on heuristics (like word matching) and force it to learn true logical reasoning.

## Part 5: Data Efficiency Analysis

(5.3) The table below shows the dev accuracy achieved when training the model on varying fractions of the full SNLI training dataset.

| Fraction of Training Data | Dev Accuracy (%) |
|---------------------------|------------------|
| 1%                        | 75.68            |
| 10%                       | 83.32            |
| 25%                       | 86.06            |
| 50%                       | 88.20            |
| 100%                      | 89.92            |

The figure below illustrates the learning curve -



(5.4) At 10% of the training data, the model achieves 83.32% accuracy, compared to 89.92% at 100% data. This means the model retains 92.66% of its peak performance using only one-tenth of the available training data.

At a 1% data regime, the variance between runs is extremely high because model performance relies heavily on the quality of the specific random sample drawn. At 50%, the variance shrinks significantly because the subset is large enough to consistently represent the overall data distribution, making training much more reliable.

The knee of the curve, representing the point of diminishing returns, occurs at the 10% data mark. Beyond this point, requiring exponentially more data yields progressively smaller marginal gains in dev accuracy.

(5.5)

1. Pre-training already teaches the model general language semantics, syntax, and reasoning. Fine-tuning only needs to teach the model how to map this existing knowledge to the specific NLI classification format, a task that requires significantly less data than learning language from scratch.
2. I would recommend collecting 10% of the data. Based on the learning curve, this point is the "knee" that achieves over 83% accuracy (retaining over 92% of the peak performance). Paying ten times that amount for the remaining marginal accuracy improvements represents a poor return on investment.